6.1. Empowering Bengal's Heritage through Quality Innovation and Livelihood Upliftment in Tussar Silk Baluchari/Swarnachari Sarees

6.1.1 History

6.1.1.1 Origin and Migration

The Baluchari craft originated in Baluchar (Murshidabad), West Bengal. Its development was significantly influenced by geopolitical shifts; in 1704, Murshid Quli Khan, the first independent Nawab of Bengal, moved the capital of the Bengal Presidency from Dacca to Baluchar. This relocation drew skilled craftsmen from Dacca and Benaras to the new capital. Later, following a massive flood, these artisans shifted to Bishnupur, where the Malla dynasty stepped in to promote and sustain the skill.

6.1.1.2 Historical Timeline and Institutional Influence

- **Ancient/Early Development:** Weavers migrated to Baluchar in the early 18th century, later moving to Bishnupur under the patronage of the Malla dynasty.
- **Colonial Era:** The craft declined significantly due to rising production costs and competition from British automated textile machinery.
- **Post-Independence**: A revival began in 1956 in Bishnupur, led by Akshay Kumar Das and the Silk Khadi Seva Mandal.
- **Present**: The industry thrives in Bishnupur through evolved techniques and modern market adaptations.
- **Major Transformations**: The Baluchari saree has experienced key technical and aesthetic shifts:
- **Materials and Techniques:** Production transitioned from natural dyes to synthetic dyes, and weaving evolved from traditional handlooms to powered jacquard looms.
- **Design Motifs:** While Mughal and British era designs depicted socio-politico-cultural events, modern designs focus on mythological stories told in a narrative style.
- **The Impact of Industrialization and Policy:** Industrialization and automated machinery caused the craft's initial decline during the colonial period. Its survival and transition into the modern era were made possible by post-independence interventions and institutional support from organizations such as the Silk Khadi Seva Mandal.

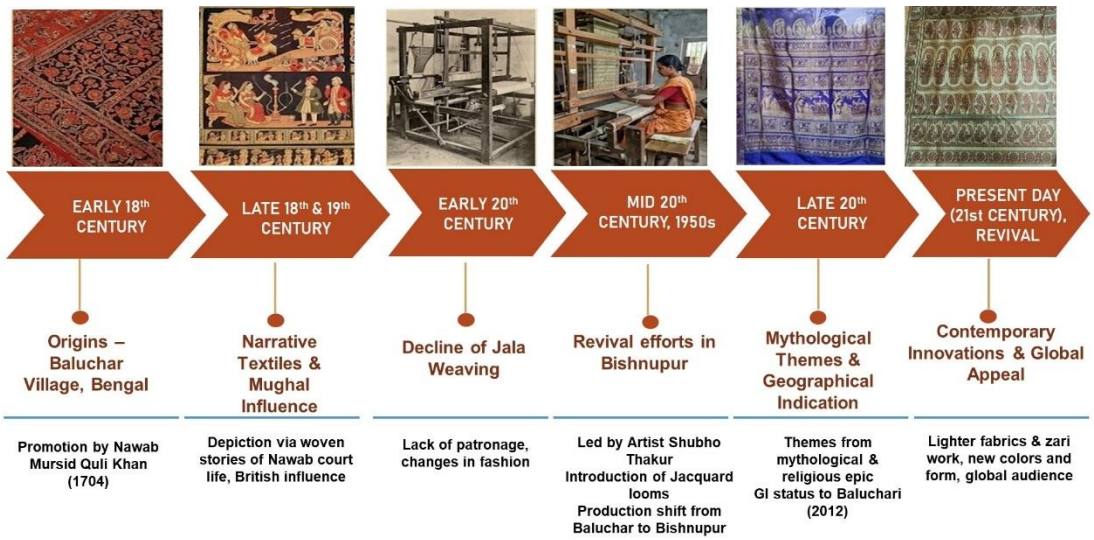

Figure Historical timeline for baluchari saree.

6.1.2. Heritage

The Baluchari sarees have been recognized as a heritage textile product for several reasons:

- These sarees are distinguished by unique motifs and patterns that make historical depictions of mythological events, geo-political issues and socio-politico-cultural events.
- Geographically, the Baluchari sarees are found only in Bishnupur, in the the state of West Bengal in India and have been issued the GI tag
- Baluchari sarees are part of the local culture of Bengal and have deep traditional significance. These sarees derive their uniqueness due to depiction of several socio-politico-cultural historical events, Indian rituals, events of mythological stories and Persian art forms.
- While Baluchari sarees have a local and national market, these are also exported to overseas customers

6.1.3 Product Features

The Baluchari saree is characterized by specific physical dimensions, structural components, and technical weaving requirements as outlined below:

- **Physical Dimensions and Weight**
- **Length:** Typically ranges between 5 m and 6.5 m.
- **Width:** Measures between 116 cms and 122 cms.
- **Weight:** Generally exceeds 400gm, though the final weight is contingent upon the quantity of mulberry or tassar silk yarn utilized and the extent of the zari work.
- **Structural Composition and Appearance:** The saree is composed of three distinct parts: the body, the border, and the pallu (anchal).

- **Visuals:** It features a shiny appearance with borders traditionally rendered in bright and/or dark vibrant colors.
- **Detailing:** Both the pallu and the border contain rich zari work, which is more concentrated in these sections than in the body of the saree.
- **Technical Manufacturing:** The weaving process utilizes two high-capacity Jacquards simultaneously, with one specifically dedicated to the ground of the textile.

Figure 3 Baluchari saree photos depicting mythological figures.

6.1.4 Cluster Information

The Baluchari sarees are made in the Bishnupur cluster of West Bengal. A detailed description of this cluster has been included below:

6.1.4.1 Workforce Demographics and Participation

Currently, approximately 12,000 to 15,000 individuals are directly involved in Baluchari weaving and its allied processes. The workforce is characterized by distinct gender and age distributions:

- Gender Roles: Men comprise 60–70% of the workforce, primarily handling loom operation and design execution. Women account for 30–40%, focusing on pre-loom activities, reeling, dyeing, and ancillary tasks.
- Age and Skill Levels: Majority (Above 40 years): Represents the bulk of highly skilled weavers: (a)20–40 years: Shows moderate participation levels, (b)Below 25 years: Reflects a critical decline in interest among the younger generation.

- Skill Hierarchy**:** Master Weavers (~10–15%): Responsible for design development, jacquard programming, and motif creation: (a) Skilled Weavers (~50–60%): Tasked with operating the looms and producing the sarees, (b) Helpers/Ancillary Workers (~20–30%): Handle silk preparation, dyeing, and finishing.

6.1.4.2 Economic Status and Livelihood Dependency

The economic landscape of the craft has shifted significantly over the last 10 to 20 years, moving from independent weavers to a system of wage labor under master weavers and traders.

- Income and Wages: The current wage for one saree is ₹1,800, which is typically divided between two artisans. This results in a meager ₹300 per artisan per unit—a stark contrast to the market price of the sarees, which range from ₹5,000 to ₹30,000.
- Dependency: While weaving remains the primary occupation for 70–80% of households, it is a secondary occupation for the remaining 20–30%.
- Diversification: Due to low wages, families are increasingly diversifying into agriculture, daily wage labor, and small grocery shops.

6.1.4.3 Challenges and Workforce Decline

The workforce is shrinking due to several systemic and physical pressures:

- Market Competition: Handlooms face severe pressure from power looms and machine-made imitations.
- Production Barriers: A single saree takes 7–10 days to produce, yet wages remain low despite the high market value.
- Structural Issues: The decline of cooperative societies and a heavy dependence on middlemen have limited direct market access.
- Health Hazards: Artisans frequently report health issues related to the eyes and the lower abdomen.
- Migration: Weavers are increasingly moving to cities like Kolkata and Delhi to pursue better-paying opportunities in construction or the service sector.

6.1.4.4 Community Aspirations and Government Support

- The weaving community seeks a structural overhaul to ensure the craft's survival. Key expectations include:
- Financial Reforms: Fair wages, profit-sharing models, and the revival of cooperative societies.
- Direct Access: Opportunities to work directly with designers and gain access to export markets.
- State Intervention: Government assistance is sought specifically for marketing, branding, and digital platforms to bypass middlemen and reach consumers directly.

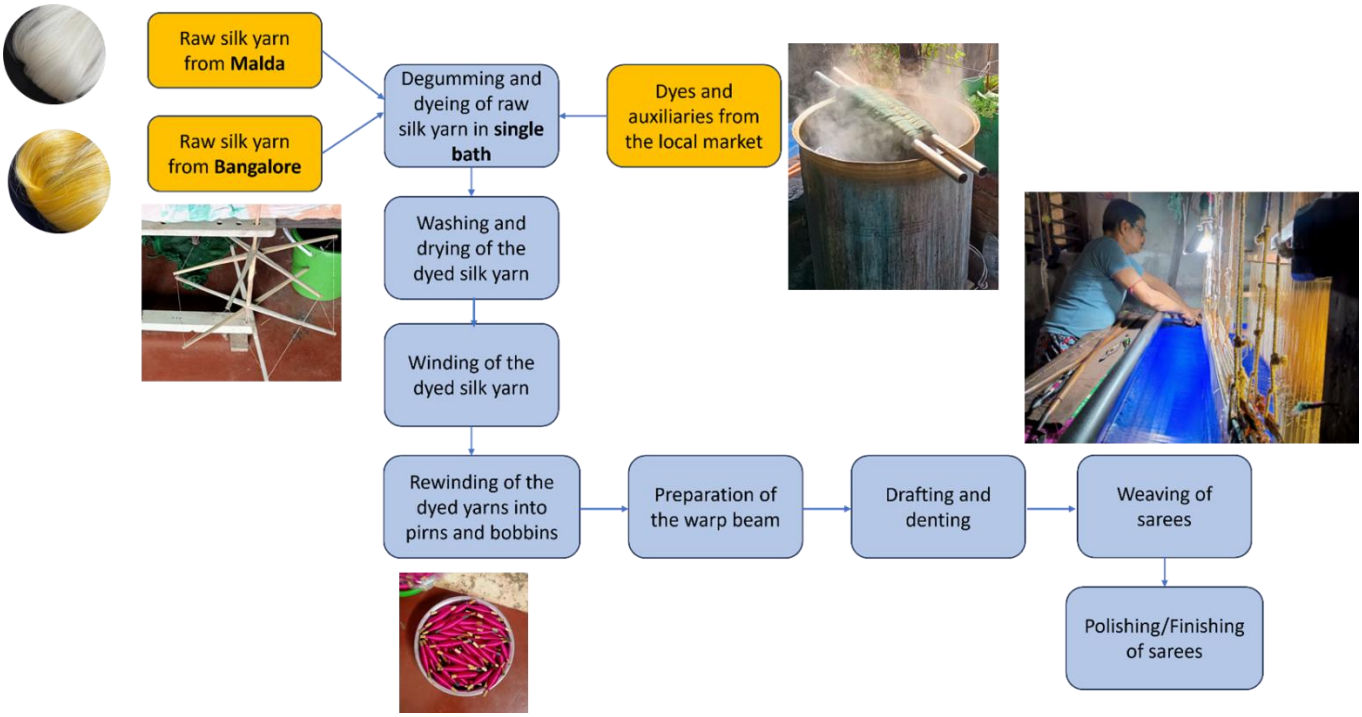

Figure Production process of the Baluchari saree in cluster.

6.2 Technical inputs in handholding muslin weavers for reviving its lost glory

6.2.1 Introduction

Muslin, once celebrated as the “woven wind” or “fabric of the wind” for its extraordinary fineness, lightness, transparency, and breathability, represents one of the greatest textile heritages of the Indian subcontinent, particularly Bengal. Muslin cloth is a plain-weave cotton fabric.

Figure 4 Different Muslin Fabric production zones of West Bengal, India

6.2.2 History

The history of muslin is defined by its ancient origins in the Bengal subcontinent, its peak as a prestigious Mughal-era textile, and its eventual decline under British colonial rule.

6.2.2.1 Historical development

The historical development of Muslin in West Bengal has been outlined here:

- Muslin originated around 1000 BCE in the Bengal subcontinent.
- Production centered in Sonargaon (Dhaka) and its surroundings, alongside Santipur and Murshidabad in West Bengal.
- It reached its peak during the 17th–18th century Mughal period.
- British colonial rule caused its decline through cheap English imports and colonialist oppression.

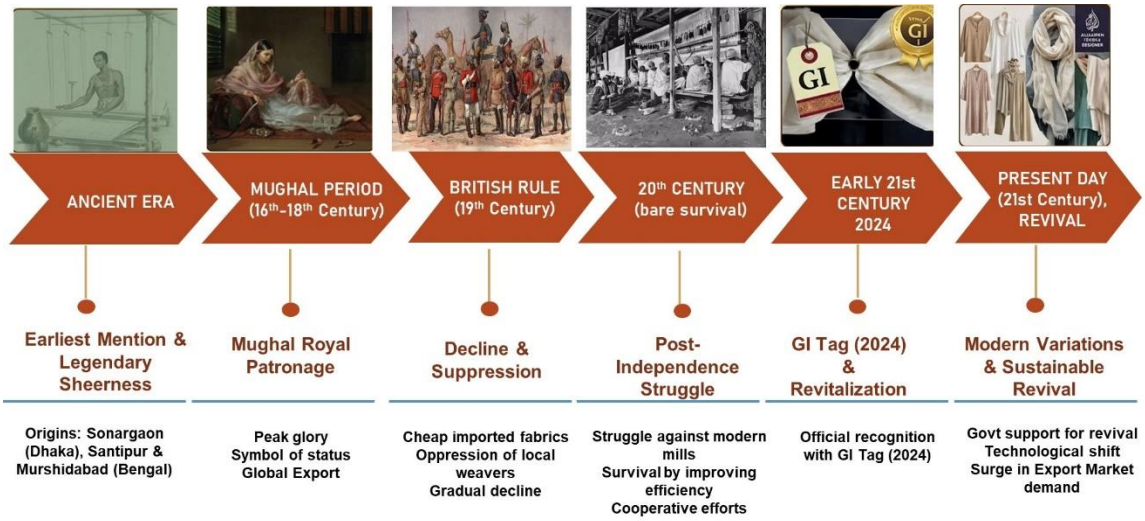

Figure Historical timeline of muslin cloth.

6.2.2.2. Major Transformations over time

The evolution of muslin production has shifted from ancient manual yarn preparation by young artisans to modern solar-powered spinning, while maintaining strictly motor-free handloom weaving to preserve its heritage and GI status.

- In the ancient times, girls in the age group of 15 to 20 prepared muslin yarn by hand without machinery intervention.
- The total beauty of the craft depended on the skill and softness of the hand.
- There was familiarity with the Ambar Charka for muslin yarn spinning in 1980.

- Over time, machine settings and parameters have changed for better processing and production.
- In modern days, solar power and DC motor-driven systems are utilized in the muslin yarn spinning system.

6.2.3 Heritage

Muslin is a globally renowned heritage textile from Bengal, distinguished by its delicate craftsmanship, historical prominence as a major Mughal-era export, and its modern revival recognized through GI status. The key reasons due to which Muslin cloth is regarded as a heritage textile has been listed below:

- Exceptional Craftsmanship: Muslin represents traditional weaving skills and local craftsmanship, along with purity and simplicity. The fabric is produced through the excellent craftsmanship of Bengal artisans, resulting in a soft, lightweight, fine, and delicate material. It is a symbol of regional cultural identity and the pride of India.
- Historical Global Prominence: Mughal Bengal emerged as the world's foremost muslin exporter, marking its historical significance during the 17th and 18th centuries. Muslim symbolised elegance, refinement, and an elite lifestyle and was integral to the royal courts of the Mughal Empire.
- Resilience and Revival: After weaving ceased in the late 18th century due to colonial oppression and English imports, successful revival initiatives were launched in India (late 20th century) and Bangladesh (early 21st century). Bengal Muslin received official GI recognition for the Indian state of West Bengal in 2024.

6.2.4 Product Features

Muslin by its extraordinary fineness, lightness, transparency, and breathability. Muslin cloth is a plain-weave cotton fabric. A minimum yarn count of 300 was used for muslin, making it soft and transparent. It is a very fine, lightweight, and comfortable fabric that can even pass through a metal ring. Listed below are the key characteristics of the Muslin fabric:

- Produced from 100 % Cotton fibre having staple length more than 40 mm
- Muslin is a fine fabric that a whole fabric can pass through a ring
- Very few fibres (5-10) in yarn cross-section as Charka based hand spinning process
- Hand Ginning and Weaving in Handloom
- Generally, no signature motifs or patters are used in Muslin
- Mostly produced as plain(1/1) in weave and white in color
- Shirt, Saree, Dhuti, Kurta- Kurti are generally produced from muslin fabric as an end product.

Figure 5 Muslin fabric

6.2.5 Cluster Information

The muslin weaving cluster in Bengal is a growing community of 500 registered artisans, primarily women, who maintain this ancient heritage despite significant economic, ergonomic, and production challenges. The details of the Muslin cluster in West Bengal have been included below:

6.2.5.1 Workforce and Demographics

- Current Participation: There are currently 500 registered Muslin weavers in Bengal (registered by MSME and Khadi).
- Historical Comparison: Twenty years ago, there were approximately 100–150 registered weavers. The workforce has been increasing by about 5% to 10% annually.
- Gender and Age: The profession is normally associated with women in the 30 to 60 age group, as soft and expert hands are required for spinning Muslin yarn.
- Skill Levels: There are only 10 to 12 master trainers, indicating a significant lack of master trainers in this area.

6.2.5.2 Economics and Livelihood

- Income: The average monthly income is normally Rs. 6,000 to Rs. 7,000.

- Government Incentives: Additional support includes a 10% Market Development Allowance (MDA) from the West Bengal Government, a 10% ex-gratia from owners annually, and a 10% MMDA from the Central Government.
- Total Earnings: With Government support, the total income reaches Rs. 9,000 to Rs. 10,000 per month.
- Dependency: Households are 50% dependent on this craft for their livelihood.

6.2.5.3 Operational Challenges and Workforce Trends

- Production Constraints: Weaving efficiency is very low due to the fabric's delicacy; preparing one saree can take 5 to 6 months.
- Time Limitations: Weaving is restricted to the hours between 9 PM and 9 AM. Attempting to weave beyond 9 AM results in frequent yarn breakage.
- Wages and Labor: Daily wages for spinners and weavers are low, approximately Rs. 200 for 8 hours of work, creating pressure to meet basic life needs.
- Ergonomics and Health: Ergonomical problems for spinners are a major issue, leading to a loss of interest in the profession.
- Migration: Skilled labourers are moving to other professions due to a lack of profit, low production rates, and the physical difficulty of spinning and weaving.

6.2.5.4 Market and Institutional Challenges

- Demand: High product prices mean demand among a broad range of customers is not yet up to the mark.
- Training: There is an insufficient number of awareness and training programs for weavers and spinners.

6.2.5.5 Community Expectations and Aspirations

- Government Support: The community seeks various forms of support from the government.
- Technological Upgradation: A need for technical support to upgrade production technology.
- Skill Development: Aspirations for different training programs to enhance skill development.

Figure manufacturing process of Muslin cloth in cluster.

6.3 Negamam Sarees: Reviving Heritage through Sustainable Innovation

6.3.1 Introduction

Negamam Cotton Sarees are handloom textiles produced in the Coimbatore region of Tamil Nadu, supported by moisture-laden winds from the Western Ghats. Typically woven from 80s combed cotton, the fabric is dense, durable, and comfortable for various climates. Production concentrates between June and November when favorable humidity aids yarn handling.

6.3.2 History

6.3.2.1 Origin

Negamam Cotton Sarees originated from Negamam village, Coimbatore. Developing after 1871, the village became a weaving and trade center linked to the Devanga community. These skilled weavers migrated from the Kongu belt, and the sarees eventually became identified by the village name.

6.3.2.2 Historical Development

Originally a local hub supplying coarse 20s and 30s cotton sarees, the region shifted to finer 40s, 60s, and 80s counts. Devanga weavers improved fabric quality and strength. From 1942, cooperative societies organized production, provided yarn, and facilitated market access.

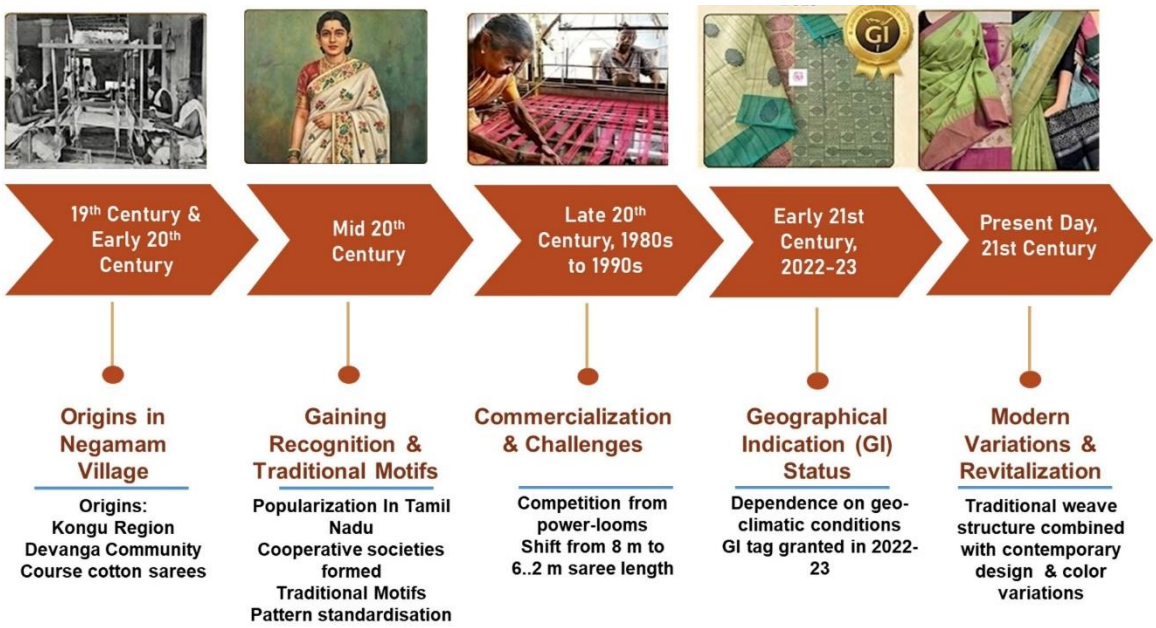

Figure Historical timeline for negamam saree.

6.3.3.3 Major Transformations

The Major transformations of the Negamam sarees over time have been listed below:

- Dimensions: Transitioned from traditional 8-meter lengths to the modern 6.2-meter standard.
- Thread Density: Shifted from 86–90 PPI to contemporary 70–76 PPI, while maintaining 80s combed cotton as the benchmark.
- Design: Simple dobby geometric patterns and narrow borders evolved into complex Jacquard layouts with elaborate motifs and high-contrast colors.

6.3.3 Heritage

Negamam Cotton Sarees are recognized as a textile heritage for their 200-year history in the Pollachi region and their deep connection to the multi-century weaving lineage of the Devanga community.

- Generational Continuity: The craft has been sustained for over 200 years, with knowledge and skills transmitted within households across two to three generations or more.
- Traditional Techniques: Production relies on time-tested, manual processes including warping, sizing, and handloom weaving preserved with minimal technological intervention.
- Environmental and Cultural Ties: The craft is supported by the local Pollachi climate and maintains strong ties to regional cultural practices and everyday life.
- Heritage Characteristics: Defined by durable textures, simple designs, and nature-inspired motifs, these sarees embody both material and intangible heritage.

- Identity: The textiles represent the regional identity and the socio-cultural fabric of the weaving community.

6.3.4 Product Features

A key feature is the richly woven pallu, featuring traditional motifs like peacocks, parrots, paisleys, elephants, and swans in repeated or mirrored patterns. Jacquard techniques achieve these details, sometimes incorporating half-fine zari for subtle richness. The key product features of Negamam sarees have been listed below:

- Material: 80s combed cotton (warp/weft); Zari or mercerized cotton for extra-weft designs.
- Technique: Exclusively handloom (pit/raised-pit looms) using traditional "Thandi" attachments for high density, natural starches for sizing, and vat dyeing.
- Motifs: Traditional icons like the peacock (Annam), Rudraksha, and various Temple (Komai) borders.
- End-use: Historically an 8-meter drape for Kongu women; currently a 6.2-meter standard saree for daily and functional wear.

Figure 6 Photos of Negamam sarees with different designs

6.3.5 Cluster Information

The Negamam cotton saree cluster is a resilient handloom community, primarily sustained by an aging workforce of Devanga weaving families who face significant economic and physical

challenges despite their deep-rooted generational expertise. The salient aspects of the Negamam cluster in the Coimbatore region has been included below:

6.3.5.1 Current Participation

The cluster has over 4,500 looms approximately, across multiple villages, involving high household participation in weaving and pre-loom processes.

6.3.5.2 Historical Comparison

The workforce has contracted over the last 10–20 years. Once the region's primary economic driver, active looms have declined as industrialization in Pollachi and Coimbatore offers alternative employment.

6.3.5.3 Demographics

- Age: Heavily skewed toward an aging workforce (40 to 80 years).
- Gender: A family-integrated effort; men typically manage high-tension loom operations and jacquard setups, while women handle weaving and critical pre-loom processes like pirn winding and warping.
- Generational Gap: Minimal participation from the younger generation (<35 years), who are migrating toward urban professional roles.
- Skill Levels: Includes master weavers, dyers, and designers. Older artisans possess high expertise, while the newer generation lacks comparable skill depth, creating a need for knowledge transfer and exploration of new techniques.
- Household Income: Approximately Rs. 20,000 to 25,000 per month.
- Livelihood Dependency: Active weaving families are 90–100% dependent on the craft, with minimal other income sources.
- Workforce Decline Factors: Long daily hours, high manual labor input, low wage realization relative to skill, and income instability due to powerloom competition and market fluctuations.
- Migration: Younger members are gradually migrating to urban centers for stable, higher-paying jobs and education.

6.3.5.4 Key Challenges

- Rising raw material costs reducing profit margins.
- Underpaid/unacknowledged family labor in pre-processing.
- Physical strain and health issues among elderly artisans.
- Gruelling daily regimes (often starting at 5 a.m.) involving sustained physical torque.
- Informal knowledge transfer facing competition from more stable, less demanding jobs.

6.3.5.5 Community Aspirations

- Stable income and reduced physical strain through improved tools.

- Greater recognition and respect for the craft.
- Easier access to loans, subsidies, and financial support.
- Steady demand and regular work throughout the year

Figure Production process of negamam saree in cluster

6.4 A Textile Science and Engineering Approach to Documentation, Preservation, and Innovation of Traditional Embroidery Techniques for Phulkari Designs

6.4.1 Introduction

Phulkari, particularly in the form of a dupatta (headscarf) or chadar (ceremonial shawl), is a traditional hand-embroidered art form from Punjab. Commonly worn on special occasions such as weddings, festivals, and religious ceremonies, it is admired for its vibrant floral motifs and intricate stitching techniques. Often described as “flower work,” it reflects patterns of flowers, leaves, and geometric designs that define its unique aesthetic. Passed down through generations, this art form represents both creative expression and shared cultural heritage. It holds deep cultural, social, and economic significance in the region, symbolizing community traditions while showcasing local beliefs and practices. Additionally, it continues to provide a valuable source of livelihood and financial support for many families.

6.4.2 History

Though the exact origin of Phulkari is unknown, it is believed that it shows similarity with various forms of embroidery from Pakistan, Afghanistan, Persia etc. Etymologically, the word *Phulkari* has been traced to the Persian words *Phul* (flower) and *Kari* (work). Similarly, *Gulkari* (*Gul* meaning flower and *Kari* indicates work) is a floral embroidery from Persia that shows distinct resemblance to Phulkari. Within India, this embroidery shows resemblance with many other embroideries like *Kasuti* of Karnataka, *Heer Bharat* and *Soof* of Gujarat.

- Origins from the 15th century with influences from Persia, Afghanistan and present-day Punjab
- Importance of women practicing embroidery finds mention in the recitations of Guru Nanak Dev (A.D. 1469- 1538), the founder of Sikhism

- Pinnacle of glory during the reign of Maharaja Ranjit Singh and coinciding with Mughal rule in India
- Originally practiced by the Jat Sikh women of Majha, Malwa, and Doaba as well as the Hindu and Muslim women of the then undivided-Punjab.

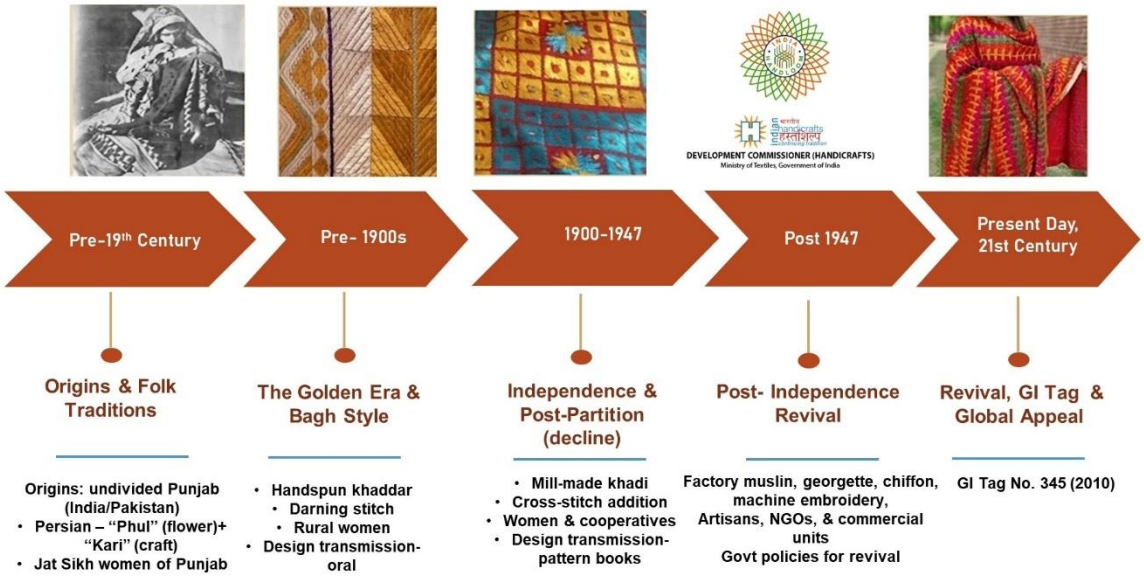

Figure Historical timeline for phulkari design.

6.4.3 Heritage

Phulkari is classified as a textile heritage due to its deep-rooted cultural and historical significance. It has ancient origins, with the craft traditionally passed down from mother to daughter, preserving skills and knowledge across generations. Information on the heritage information of Phulkari designs have been discussed below:

6.4.3.1 Historical continuity and Cultural significance

- The embroidery serves as a vivid expression of community life, and rural lifestyle of Punjab
- Depiction of beliefs, values, and emotions through intricate embroidery
- Living cultural institution of Punjabi womanhood.
- Mothers and grandmothers begin embroidering daughter's trousseau from the time of her birth
- Often stitching prayers, blessings, and life narratives are embroidered onto every thread

6.4.3.2 Traditional craftsmanship & materials

- Phulkari is a handcrafted textile art and is created by hand using silk floss (pat) on handwoven cotton fabric
- Traditionally characterised by a darning stitch worked from the reverse side of handwoven khaddar fabric

6.4.3.2 Geographical linkage, Recognition or protection status

Phulkari designs were traditionally produced by the Jat Sikh women of Majha, Malwa, and Doaba sub-regions and also by Hindu and Muslim women of the region in undivided-Punjab. Before the Partition of India, important centers of Phulkari embroidery were spread across districts such as Amritsar, Ludhiana, Jalandhar, and Patiala in India, as well as Lahore, Rawalpindi, Multan, and Sialkot in Pakistan. This shows that Phulkari belongs to the wider Punjabi cultural belt rather than being confined to modern political boundaries. In recognition of its geographical connection to Punjab, Phulkari designs was granted GI tag in the year 2010.

6.4.3.3 Continuity of practice (living tradition)

This craft has been traditionally passed down from mother to daughter, preserving skills and knowledge across generations. Traditionally women practiced this art as a leisure activity.

6.4.4 Product Features

Phulkari involves surface embroidery using vibrant floss silk (pat silk) thread worked from the wrong side of coarse hand spun khaddar or khadi cloth, creating on the right side. geometric, floral, and figurative patterns. The various product features of Phulkari designs with a focus on materials used, embroidery technique, characteristic motifs and end-use of this craft have been listed below:

6.4.4.1 Material

- Traditionally made on handwoven cotton fabric known as *khaddar*, providing a coarse yet durable base
- The embroidery done using brightly colored silk floss thread called *pat*
- Satin-like finish on the front

6.4.4.2 Technique

- Hand-embroidered using a distinctive darning stitch
- Embroidery on reverse side of the fabric to create a smooth
- Precise and intricate patterns created without pre-drawn designs
- Final product is characterised by geometric precision and textured appearance.

6.4.4.3 Motifs

Vibrant and symbolic motifs inspired by nature and daily life including flowers, leaves, crops, animals, and geometric patterns. The term “phulkari” itself means “flower work,” reflecting the dominance of floral designs. These motifs often carry cultural and emotional meanings, representing prosperity, fertility, and spirituality.

Figure 8 Photographs of Phulkari designs

6.4.4.4 End-use

Traditionally, phulkari is used in garments such as dupattas and shawls, especially for weddings, festivals, and religious ceremonies. It is an essential part of bridal trousseaus. In contemporary contexts, it is also adapted into a variety of products like sarees, suits, stoles, home furnishings, and accessories, blending traditional craftsmanship with modern utility.

6.4.5 Cluster Information

Phulkari clusters are spread across various regions in modern day Punjab. A few details of the manpower and Phulkari cluster information and enclosed below:

- Total registered Phulkari artisans (DC Handicrafts): 33732
- Estimated unregistered / informal practitioners: ~ 65,000 – 80,000
- Artisan clusters (primary): Amritsar, Patiala, Gurdaspur, Hoshiarpur, Rupnagar, Sri Muktsar Sahib, Tarn Taran, Shahid Bhagat Singh Nagar

The key occupational health concerns faced by artisans working on Phulkari designs have been recorded in the chart below:

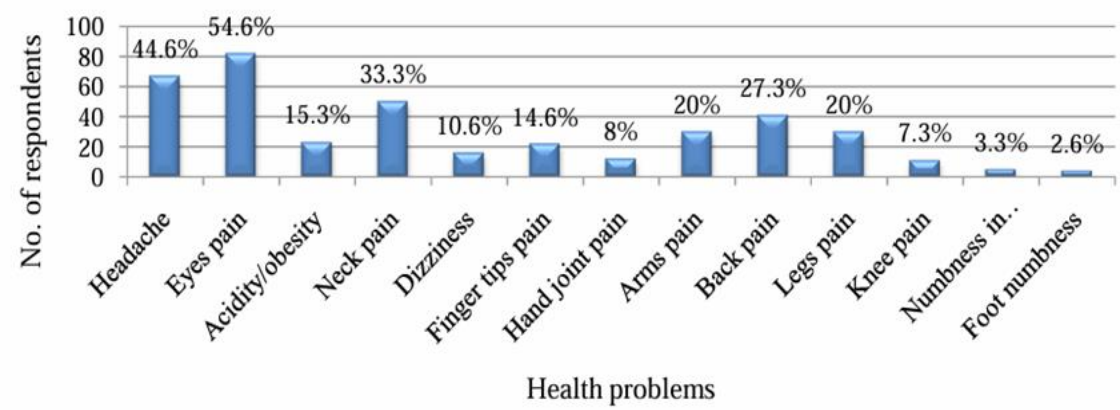

Figure 7 Health issues of Phulkari Artisans

Figure process to make phulkari design in cluster